

Smart Irrigation System

Presentation Script — Pointer Format

One line per point, not a script to memorize. Know your section, hit the beats, hand off clean.

SPEAKER 1 OF 4

Sohan

Introduction — Slides 1, 2, 3

SL-01 Title

- Greet the room — good morning / afternoon
- Project: the **Smart Irrigation System**
- Two parts: a working circuit built in Tinkercad, and an interactive 3D digital twin built in Three.js
- “Let’s walk you through both”

SL-02 Project Overview

- **Step 1** — senses soil moisture + temperature (soil sensor, temp sensor)
- **Step 2** — potentiometer sets a custom threshold, how dry before watering starts
- **Step 3** — threshold crossed → pump, LEDs and buzzer switch on automatically
- **Step 4** — live readout on the real LCD and on the 3D screen

SL-03 System Architecture

- Arduino Uno = the brain of the system
- Inputs: soil moisture sensor, temperature sensor, threshold dial
- Outputs: water pump, LCD display, buzzer + LEDs
- Arduino constantly reads inputs and decides the action

→ hand off to **Manthan**

Manthan

Tinkercad Circuit — Slides 4, 5, 6

SL-04 Part 1 Divider

- Moving into Part One — how the circuit was designed and simulated in Tinkercad
- All done before touching real hardware

SL-05 The Circuit

- Full circuit, simulated entirely in Tinkercad
- Arduino Uno as controller
- Soil moisture sensor + TMP36 temperature sensor
- DC motor standing in as the water pump
- 16×2 LCD for readouts, potentiometer for the threshold, buzzer + LEDs for alerts
- Wiring tested and verified in sim — caught mistakes before building anything physical

SL-06 Bill of Materials

- Full parts list — every sensor, resistor and component, with quantities
- Doubles as the shopping list and the wiring reference

→ hand off to **Aayush**

Aayush

Logic + 3D Intro — Slides 7, 8, 9

SL-07 How the Circuit Works

- **Step 1** — sensors read soil and temperature
- **Step 2** — Arduino compares the reading to the set threshold
- **Step 3** — decide and act: too dry → pump, red LED and buzzer on; fine → green LED stays on, system idles
- **Step 4** — everything shown live on the LCD

SL-08 Part 2 Divider

- That's the hardware side covered
- Part Two — the same logic rebuilt as a 3D digital twin in Three.js

SL-09 Inside the Scene

- Full 3D farm scene, runs in the browser — no install needed
- Sky with weather effects, water tank, windmill, shed, crop fields
- Live HUD/dashboard — moisture, temperature, rain, tank level, system state
- Every texture — grass, soil, wood — is generated by code, not an image file

→ hand off to **Naman**

SPEAKER 4 OF 4

Naman

Features + Wrap-Up — Slides 10, 11, 12, 13

SL-10 Play With It

- The 3D model is fully interactive
- Drag the on-screen dial — watering threshold changes live
- Click a button to simulate dry soil, watch the system respond
- Trigger rain, toggle day/night
- Orbit the camera around the whole farm by dragging the mouse

SL-11 Two Builds, One System

- Same system, built two different ways
- Tinkercad — real, hardware-accurate circuit; great for testing and validating before buying parts
- Three.js — digital twin; runs in any browser, zero install, built for demos exactly like this one

SL-12 Impact & What's Next

- Shows hardware prototyping and software visualization working as one workflow
- Next: stream live sensor data from a real board into the 3D dashboard
- Make the system solar-powered for field use
- Add a mobile app for remote control
- Expand to manage multiple irrigation zones at once

SL-13 Thank You

- Thank the audience for listening
- Open the floor for questions